

ANUJ DUBEY

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EDUCATION

Indian Institute of Technology(IIT) Guwahati, India

B.Tech. in Electronics & Communication Engineering | Minor in Computer Science & Engineering

Jul 2022 – May 2026

CGPA: 8.38/10.0

EXPERIENCE

Hybrid GNN-Augmented LPA* for Urban Traffic Networks

Research Intern | Prof. Anirban Das Gupta, IIT Guwahati

May 2025 – Aug 2025

Guwahati, India

- Created a **hybrid GNN** combining STGCN and a custom FDGNN for dynamic traffic prediction with uncertainty quantification, achieving **22% lower MAE (3.21 km/h)** than DCRNN and **95.8% calibrated confidence** on METR-LA.
- Designed a **risk-sensitive cost function** $\mu + z \cdot \sigma$ integrated with GNN predictions, enabling uncertainty-aware **route optimization** in sparse urban graphs and reducing delays in dynamic traffic scenarios.
- Optimized Lifelong Planning A* (LPA*) with multi-scale cluster-based updates via spectral clustering, yielding **2.7× faster computation** than standard A* while preserving **93% path optimality**.

PROJECTS

Z-Coder

Coding Club, IITG

May – Jul 2024

t.ly/DieuI 

- ◇ **Technologies Used:** NodeJS | ExpressJS | HTML | PostgreSQL | CSS
- Developed a full-stack platform with problem bookmarking, code evaluation via **ZCoder Compiler**, and PostgreSQL backed **persistent state handling** across sessions for coding practice, with integrated **JWT-based** authentication.
- Built a custom cross-browser **JavaScript extension** to autocapture and **sync problem URLs** with user dashboards.

Decentralized P2P File Sharing

Group Project

Mar – May 2025

t.ly/AXErs 

- ◇ **Technologies Used:** ReactJS | NodeJS | ExpressJS | WebRTC | Socket.IO | CryptoJS
- Developed a **decentralized P2P platform** for **secure file sharing** and real-time chat using **WebRTC** and **Socket.IO**, with **AES-based** encryption, dynamic room management and **chunked file transfer** with **zero-byte transmission error** handling.
- Integrated **peer exit detection**, **room overflow alerts**, and secure message exchange via **CryptoJS** across communication.

Spilling Register Allocator

Personal Project

Aug – Sep 2025

t.ly/zJasz 

- ◇ **Technical skills Used:** C++ | Algorithms & Data Structures | Graph Theory
- Developed a **register allocator** using Iterated Register Coalescing, building an interference graph from variable liveness.
- Implemented **spill-cost model** to reduce redundant instructions and **minimized performance loss** for high register pressure.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python
- **Database Management:** MySQL, PostgreSQL
- **Web Development:** JavaScript, NodeJS, ExpressJS, ReactJS
- **Python Stack:** NumPy, Pandas, PyTorch, Matplotlib

ACHIEVEMENTS

MIT Informatics Tournament	Achieved Rank 162 in the penultimate round among 1K+ contestants.	2025
Leetcode Coding Platform	Peak rating of 1877 , earning the title of Knight . Handle: <i>Trampoline1</i>	2025
Codeforces Coding Platform	Peak rating of 1490 , earning the title of Specialist . Handle: <i>This_is_Adroit</i>	2025
Meta Hacker Cup 2025	Achieved Global Rank 611 among 12K+ contestants in Round 2 worldwide.	2025
Codeforces Round 952, Div-4	Achieved Global Rank of 295 among 40K+ active contestants worldwide.	2024
Codechef Starters 136, Div-3	Achieved Global Rank of 236 among 12K+ active contestants worldwide.	2024
Branch Change to ECE	Awarded for ranking in Top 5 among 95+ first-year students across the batch.	2023
JEE Advanced 2022	Secured All India Rank 5127 among 0.15 million candidates.	2022
JEE Main 2022	Secured All India Rank 7434 among 1 million+ candidates.	2022

KEY COURSES TAKEN

Computer Science: Data Structures & Algorithms[†], Operating Systems, Computer Networks, Software Engineering

Electronics: Control Systems[†], Digital Electronics[†], Pattern Recognition & Machine Learning, Blockchain Technology

Mathematics: Probability & Random Processes, Linear Algebra[†], Graph Theory and Applications

[†]AB+ Grade Awarded

POSITIONS OF RESPONSIBILITY

•CP-Algorithms – Open-Source Algorithms Community:

- **Contributor** : Revised and **enhanced** an existing algorithm article by improving logic and refining explanations. 2025

•Computer Science Engineering Association (CSEA), IIT Guwahati:

- **Web-Dev Mentor** : Mentoring **20+ students** in building real-world web development projects and guiding them. 2025